ActInf Livestream #034.2 ~ "The free energy principle: it’s not about what it takes..."

Source: YouTube **Playlist**: Active Inference Livestreams (Paper discussions)
Duration: 1:42:55 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

then [Music] all right hello everyone welcome to action flab live stream number 34.2 it's december 14th 2021 welcome to the actin flab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at some of the links here on this slide this is a recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams if you check out this coda link which i will post into the youtube live chat right now you can see a lot of information on past present and future live streams and also more information on if you want to participate because that's what a participatory lab is if you're hearing this you have the affordance to get involved today in active stream 34.2 we're going to be having our third discussion on the paper free energy principle it's not about what it takes it's about what took you there by axel constant from 2021 and we're just going to be having a fun dot 2 jumping off we'll go over any parts of the paper that we want to look at return to some of the formalisms and then also we've written down a few questions specifically to address but of course anything people raise or anything that people ask in the live chat we can get to that and usually it says next week we will discuss other papers or we'll be discussing the same paper but actually it's next year because this is the last regularly scheduled weekly journal discussion that we're doing and we'll have a little bit of an encore on this thursday with uh 32.3 when conor hines will join for a dot three that's kind of like the triple jump the the hop skip in the jump uh rather than just the mirror jump but this is kind of the ending of a really awesome year of discussions and we haven't quite set which papers to discuss next year so if you have suggestions or you want to get involved in the dot-coms organizational unit where we make these kinds of decisions collectively then just contact us through any channel okay so we'll begin with an introductions and warm-up section so people can say hello and they can address any of these warm-up questions like what they're excited to talk about today or just anything else they want to mention so this should be fun and thanks to everybody who's joining on the video stream and those who are watching live on youtube so i'll start i'm daniel i'm a researcher in california i think today i'm a little bit excited about talking about how we got here where we are and then where we go what is the free energy principle and active inference say about where we go and i'll pass to blue hi i'm blue i'm an independent research consultant in new mexico and i am excited to talk about um the difference between living and non-living systems and where the fep may come into play there all right how about dean sorry i actually don't hear dean hi sorry about that i'm dean i'm obviously not fully awake and i'm still warming up because i'm in canada in the mountains and it's snowing as we speak um what's interesting to me about this paper and about its role in terms of fep and active inference is so once we have identified the entailment problem then what happens and i'll pass it over to stephen hello steven here i'm in toronto and i'm interested in this reframing um i think this paper does i think it's helpful um maybe thinking about you know where was i in the river of life when something happened to take me to where i am rather it sort of might be never fully known but the principles might be useful i like that rather than this is the calculation of what happened which is that temptation so i think there's something about that going on here so look forward to pursuing that more i'm going to pass this over to dave the two things i'm excited about are that in this city north of manila which has rougher the population of pittsburgh or anaheim or manchester we had one new case of coven yesterday and zero today and also as soon as daniel pulled his pope alexander the sixth routine and said when will you make an end suddenly maria and i are just barreling through the redaction of the seminar who who is there anybody that's all yeah dave was just mentioning there the um imminent completion of the firstin symposium transcript where we're taking the symposium that we had earlier in the year and enriching it into something that can be used for closed captioning as well as translations and enrichment and connections to other aspects of our knowledge base so that will be very fun and it's also speaks to how we got here and where we're going i think because eventually we'll give that kind of a treatment to all of these live streams that will be quite interesting so steven yes please i'm actually so because we're using a slightly different jitsi um let's just get all of our emojis out of the way if you go to the hand and then you do emojis it does like okay oh perfect disable sounds for all please disable it okay now feel free to use emojis you can signal by repeatedly clicking how shocked you are for example at what dean says okay stephen please yes this actually it works a bit more cleanly as well i think this this interface um i was gonna say that i'm interested as well um the the question of scale the question of scale i think there's something in this is like we're saying um it's not how you got there well what scale is you is it scale free is it scale friendly i know we've been talking about that but is is there um is this is this you um like the royal one is it like a very broad term or is it something that's like very much um at the scale of an agent is it across scales nested scales um i'm not saying this necessarily goes into that but i think because i'm interested in okay taking this philosophical underpinning which at first looks like it adds an extra burden or becomes a side issue for a more applied context actually i think it it does help just back up a bit it helps back up and in doing that backing up i think it keeps us with that unifying principle um approach rather than a unified theory temptation which i think is what often happens to everyone right we so i think that keeping it at that slightly more nascent understanding may give us a way to like understand the types of principles that could be seen to have played a part which could therefore be seen to have been useful in other people trying to do something similar which is not the same as a unified theory of why something happened thanks steven dean what the paper brings out for me and and it kind of ties into what stephen was just saying is that you have an entailment question that you want to have answered and now you have to think of that in terms of what's necessary and what's sufficient if you look at that uni-directionally meaning through the course of time as the flow or the river or whatever metaphor you want to apply here if you look at that one directionally that's going to kick out one version and what this paper points out is that there's two directions at the same time so there's a relationship between meaning and a different form of entailment not one causing the other necessarily but the fact that there are two and now they are somehow relating to one another which is a which is a different animal and so i think what i'm going to be really curious about in terms of the point two of this of what we're doing here today is seeing whether or not we can address what each one of those things one unilater unidirectionally through the course of time and the other strictly relationally in terms of how to how do two things that we know are present interact with one another i'm going to be really curious to see what other people think might be the difference between those two very two different views because i think that's what the author asks us to entertain is not just the one but both cool i think to to steven's question about scale i think it's also a important thing to keep in mind so the example in the paper which we talked about in the dot zero and not one was this bacterium so this is the toy example that shows that first off the differences between the bayesian bacterium and the variational bayesian free energy minimizing bacterium and then shows how it's not simply sufficient to be minimizing free energy but rather one has to be minimizing free energy with the appropriate priors so it's not enough to be doing just bayesian statistics you have to have the right priors in order to make the right kinds of decisions and then the relationship between free energy minimization and life is what is discussed as the entailment problem so we talked about those earlier so just go back and check if you'd like but what is the scale at which the entailment problem is even applying is it the cell membrane of the bacterium it just aligns perfectly with the physical scope of the agent or is it the colony of bacteria or is it the bacteria and its niche that has to be explained how it got there aren't we always going to have supra systems that are going to need their own explanation and i think that really speaks to also what you said stephen about the difference between the unifying approach versus the unified theory this is like past tense it's been unified what else do we need to know there's only one versus unifying is this ongoing process and it's an approach where we might have some guidebooks or tools memes for the road but it's not over yet we're still approaching some other question but then how did we get that other question yes stephen and also i don't think that people need to raise their hand maybe they can throw up an emoji silently because it still blinks a little bit but i think if you just use for example laughs then i'll i'll see a flurry of laughter oh nice then i'll be laughing as well that's not probably a bad thing make me a bit more cheerful um yeah i've got this vision of these little faces now um there's um there's something i'd be interested in a bit more on what you were just saying there basically because you you said um there needs to be more um i'd be interested what people's thoughts are around what makes this helpful from an applied perspective so because one of the big so i think one of the key things often in in a more applied perspective is um there's maybe the application of some kind of minimal model theory that's been shown to give a type of transitional structuring for instance what some cognitive agent who might be engaging with the world more like someone with autism the nature of how they process sensory information or other types of situations so that kind of general sense of the types of ways can then be used and maybe applied at a bigger scale but generally speaking you also once you get if you if the applied side of it is starting to go beyond a fairly low dimensional model and into these more complex real life that's not the right term but applied context these heuristics might be more broad like the unifying approach is is going to be maybe more valid and i'm wondering how does the jump from a biological philosophy mindset that's informing this paper translate into those so i'd be curious just what others thoughts are on that good question how does philosophy impact or influence application how do basic or theoretical biology questions influence the day-to-day and the way that different projects are carried out because it seems almost um trivial to say that the projects that we're implementing whether it's a road construction or whatever it is it kind of has a biological component it's by us for us and then it has a philosophical component even if only implicitly in the top-down structuring so we're talking about philosophy and biology i think that's even what journal the paper is in so then what does that mean for how we apply and yeah what else is part of the approach i think these are interesting questions yeah go for it one thing i've been very interested in there's a new field by michael quinn patton around principles focused evaluation which sort of extends his utility focused evaluation and developmental evaluation where you you know your your things are unfolding so the evaluator is not it's not evaluating after the fact but is it part of the strategic journey to see well what what sort of um you could say in our case what sort of free energy like minimizing incidents happened that looks significant to the context okay so it's it's it's not completely noisy it's not completely unknown yet it's complex enough that we're not following um a rolled out system that's kind of relatively dead and relatively um slapped down on the table each time like a piece of meat that's dead it's something that's kind of evolving and going and i think there's something quite nice here to that principles approach um you know you you and what principles are going to be useful when you're looking at understanding if there were particular choices in the particular stream that happened to have been present that made free energy minimizing happen or maybe when it didn't happen when the agent didn't manage to make the choices and things started to unravel that could also be interesting because often it's more interesting to see you don't really know which one made the difference when something has been kept on track because it it never fully uh you know spiraled out and magnified but what were the times when suddenly went wrong and or can we find out trace where that perturbation that started to rock other parts of the system of uh interest came from and that that might be somewhere this type of work ties in as a thought daniel can i can i ask a question without raising my hand yeah i think we can just go for it in this one so here's my question and it ties in directly with this paper i don't think the author is asking in in the generic sense which came first the chicken or the egg any more than i think he's when he's looking at an entailment problem which came first sensing or acting which came first internal or external which came first um in all aspects of the philosophical sense of what's the relationship between internal external sensing acting i think what trips people up and what gets them on that unidirectional linear kind of approach is asking what's first instead of holding up the relationship between and then seeing what falls out the other side if it's an entailment question is internal necessary for external linearly speaking you could make that argument but i don't know that that necessarily holds then if you're saying does external then make necessary internal however you look at it as a relationship between internal and external both are necessary so one can be satisfactory for the other if you're looking at it unidirectionally but if you're looking at it bi-directionally now all of a sudden the frame shifts and both become necessary in a relationship in order for the relationship to continue forward so i think for a lot of people asking which came first the chicken or the egg from a relational standpoint doesn't matter they both have to exist for the question to continue not which came first but do both exist and i think that's the entailment question that this author is asking us at least contemplate a little bit are we are we holding ourselves to one view that unilateral view when really what we should be doing is looking at the relationship and what entails the other as being either status satisfying or necessary great stephen thought on that and so following on from dean there's maybe is like you say when we get into do they exist now we're into the structural stuff we're into the structures we're into the structures where action and interaction happens as opposed to a meta system of egg and you know there's a structure and i suppose the question can also be do they sufficiently exist and at what point so for instance at what point maybe through pollution lack of calcium and diet kinetic impact are they no longer sufficiently persistent and suddenly one of the eggs can't make another chicken okay so that that that could become you know that then maybe is out of the how did 50 000 eggs get made well maybe it's more useful so when at what point did point one percent of those eggs not continue what was then outside of pullback attractor or whatever way we want to say it that was needed to keep it within a viable um a viable living trajectory so one kind of funny answer to which came first the chicken or the egg from an instrumentalist perspective you go oh in your sentence chicken just that's the answer the way you spoke that question it was chicken and someone said well what came first the egg or the ch okay then the egg came first in how you asked it so what came first in your model but that question smacks of realism we start thinking about the the actual way that things might have played out and is it more the case that there was a chicken that that's the realism question and then when we take this relational instrumentalist perspective um there are different kinds of time relationships falling out and i don't know if these are the right words but just a few of them one thing is about the internal and external that dean mentioned the partition is simultaneous when we do a particular partition of internal external and their corresponding blanket states those are all co-instantiated at the same time in model time or in the relational instrumental time that doesn't mean that they had to have been coming into existence as actual external entities whether that static structural ontology or a dynamical process ontology but once we're in the modeler's time the partitioning comes into existence at the same time and then we have things that are changing through time so there's the intra model actin formalism like the subscript next to the letter when we're talking about stay at time one two three four that's within the model within the active formalism changes through time then there's the extra formalism like outside of the kernel of the model which is the iterated modeling which is also changing through time because it's not like any program is just produced and then that's the final version there's always an iteration process in the real world so that's changes through time somehow spinning that around we actually get to this timeless and momentary that's like the complex entailments where we start to ask about the relationship that's actually not conditioned on a given moment but not sure exactly if that's the right way to talk about that but there is something that's it's not time travel but it's not just forward in time but it's not merely reflection or memory on the past so daniel here's a here's a thing to think about if you start with the model if you begin with the model you you start it at near time zero in even in the model that axel brings out it's simple then by its place in the time continuum if you end with the model it's got to be much more sophisticated because a whole bunch of things have happened a whole bunch of new information has been available now to include in the prior so i don't know that it's it's complex entailment happens the further out you wait for the model to appear if you start you're going to get a different result than if the model is closer to the end of the loop closure that's that's what this entailment question is actually surfacing for us it's when the model so where did that come from space time means when the model when did the model appear so that we close one loop and start the the next update that's that's where the sufficiency and the necessity part gets really strange because it's it's essentially a space-time question and if we include two arrows we're now including space and time thanks dean stephen within that space and time i think as daniel has mentioned we can have these implicit assumptions just in the way we frame in our questions which uh would be subliminal but it's it's sitting there right like you say by it by even talking about things in certain ways it privileges a certain type of thinking and i i'm really interested um even in this paper we talk about what did it take for you to get here okay so there's also it's almost like that picture of like you know the beautiful puppy it's like and i can imagine asking how did it take what did it take for you to get here right but there's also those really inspiring moments when you see the really sort of starved small dog that's kind of in a really bad shape right and i was thinking what does it need to get back to there or something like that i.e a relatively viable healthy thriving type of animal so that's also and i think that often is like the priors in some ways when things have gone out of kilter there's a need there's been a prior set of experiences which have been non-optimal but hasn't killed the animal well how do those what priors need to be fed in to get back on track although the word track is not the right term but to get closer to something and even there i'm struggling i think this is where the ontology should i be saying get back to a non-equilibrium steady state to an attractor to a thriving and maybe it doesn't need to at that point maybe there's there's some concept that doesn't have to be in the free energy principle exactly it's like some idea of a thriving type of organism but that that that that situation has taken um has taken the the the living entity away from the stream that would be desired and has to find a way back or it won't survive blue i i was like just to kind of hit on both what dean and steven were saying like how did we get here like you i often think of here as like um like a physical place but it's also like to this present time point or here like how did we become alive and get to this like non-equilibrium steady state so it makes me wonder what dean was saying like here is also a when right like so what are all the things that brought you to this moment and really like i don't know like so all of the circumstances that have made you in this present moment who you are like whatever you are separated by your partition like is the present uh part of the steady state like like that that's what makes me think of that i mean the present can't be like a steady state system so it it's because it's like always constantly like going forward so i don't know it makes me think of that as an attractor is the present moment can that be an attractor like is is that possible stay in the moment stay in the present when we think about mental action like like what does that mean to stay in the moment like can that be like an attractor state like is the future an attractive state i don't know just make me think about that certainly some kind of pull from the future what's happening with free energy minimization in the physical gibbs free energy sense it's as if there's a pull from the future down the hill but it's actually this sort of instantaneous movement of the reaction yet it does act as if there's this long-term pull and also maybe the necessity and sufficiency so what does it take to get here it depends how stringent we want to be with what's here and now because even some tiny tiny gravitational factor of some other planet or some object on the other side of the planet it's safe to say that some molecule in a given body would be in a different orientation even if trivially so so then is that going to mean that everything is sufficient like we it was everything that it was there was sufficient to get us here but also everything was necessary is that too stringent or how loosely do we hold the system and i think instrumentalism gives us an answer which is just how detailed is our model and then that's the level of stringency which variables are we going to include and what precision will we have over those variables and then just to the question about thriving i guess that stephen asks whether that's in the active ontology or not so it'll be fun to find out what is and isn't in the ontology and then translations or transpositions of memes from active to apply to those other cases so like for example the person looking at the puppy and thinking here are the physiological variables that i can measure there's the shine on the code and there's the blood sugar and there's the cortisol level that's my instrumental model there's other things happening that i don't know but those are the ones and my preference for the attractor would be dot dot dot and so what policies can i take in relationship to this animal so that its attractor is now as if it's here in that three-dimensional space of shine and blood sugar and cortisol rather than the current attractor that it appears to be orbiting so then we can just talk about preferences and expectations and policies of the caretaker and then we've kind of transposed this sort of like floating obligation for the puppy to be healthy but then things do die we've transposed this kind of implicit philosophy around a floating obligation into a specific relationship and the policies that a specific caretaker in their model are going to engage in that's uh that's a good point and i think actually this is when we we find ourselves getting back into the core of what's the agent's journey in different contexts and getting into that and becomes contextual what's the expectations what's the surprise when you're into the agent's journey what was surprising or what's relevant out of all the huge amount of surprises that can be going on across all levels is kind of what becomes important potentially because that becomes where choices were being made or not been made and i think that when we talk about or when often there's this question about the minimal conditions for life that becomes a sort of false question because it's again it's in that kind of the fruit what's the minimal questions to get life now that's great someone's applied active inference work is to create life from scratch i can't see anyone doing that for a long however if we want to know what were the key expectations surprise or policy shifts in a living entity that got them to this new living entity because there's changed right i've got the scars on my hands but maybe i cut myself with a knife and i had to heal what were the what were the kind of journeys on that route and those are the things outside of a monty python foot coming down in my head in which case i don't have much surprise or because i'm gone is i'm is what were those that i think is more interesting and that is maybe what this talks about because one of the key things he talks about is structural integrity how to resist the loss of structural integrity he doesn't say how to keep the system going doesn't matter what we see the system to be it's if it's acting it's got structural integrity to be a thing with the ability to act whether that thing is now looked at as a system is academic in the discussion on the metamorphic the paper writes it may be said that the life cycle that corresponds to the thing whose integrity is maintained over time is the process integrity using bucky fuller's ontology rather than the structural integrity the structure of the egg is always being broken down the larva is always breaking down the pupa is always the adult is always breaking down and forming but there doesn't need to be structural integrity or the retention of change at any point there's a higher order uh integrity in the reproductive process and then that's the question is okay so can we just look at one nest mate do we have to look at the colony colony in the symbionts colony and the aviotic niche what is that life cycle thing what's the thing's life cycle is that gonna be like a realist answer where there's kind of one true thing whose pattern integrity is being retained might there be multiple possible answers for which thing is having its life cycle maintained there wouldn't be a answer for what is life or is are things alive it'd be model dependent and i think that's what we've one of the things that we've wrote down to explore about like what is life obviously classic questions how does the fep and actimf apply to all kinds of identified systems animate meaning with activity and motion sentient meaning with sensation inanimate without motion and an incentive without perception yeah which systems is it a slam dunk for and where is it going to be where is it overpowered where is it underpowered what changes would it need to be more adequate in those different areas can i add something daniel if i'm a rock without anthropomorphizing this if i'm a rock i can be now and now and now or i can be then and then and then but it would seem to me what the free energy principle tells us is there is some relationship to these living things because they whether they can articulate it or not they have some sense of the relationship between now and then whether then is in the past and they have a prior or then is in the future and they have a projection they're not like the non-living things which don't require some relationship between time points and i think that's one of the things that again axel is trying to point to in terms of an entailment question yes i can be in the moment and i can be mindful but the only reason why i'm aware that i'm in this moment and not in the past or then in the future is because there is something other than now and i think that that again kind of speaks to when we can take this and look at it as bi-directionality as opposed to unidirectionality just entropy carrying out what entropy does when we think about this free energy principle applying and like dean mentioned you know entropy will be doing what it does you mentioned earlier about the flickering flame okay now i've got a gas flame okay if i maintain the input of gas there will be a structural integrity of the flame over time so the flame as a at one scale with one perception of what it means to be a flame it could be seen to be persistent however does that mean that it on that is time average in trying to maintain if the lens is a flame-like structure which maybe if coupled to our niche somehow we are trying to make it do that but then becomes the question of where is the variational free energy being looked at is is it about free energy minimization or is it about leveraging the fact that there is a free energy principle happening there is free energy minimization there's but there's also just free energy variation so i could look at that and i maybe get some insight by seeing the flickering flame which isn't structurally hasn't got that much integrity but it has a form that is always changing yeah has a certain at scale of information i know i should say scale of information or something but some qualitative nature of information persists over time where is that flame in relation to is it just because there's a fluctuation that i can extract the information from the persistence and the variation even if there's no inherent motion to reduce dissipation per se interesting i think that if we return to the entailment problem and the strong and the weak claims so the strong claim minimizing free energy is sufficient for the overly generous claim and then the weakness st if the system is currently alive which as we looked at with that metamorphic slide means that it has persisted in its pattern integrity which i know it was used as structural integrity but that's actually a static concept it means that it minimized its free energy the flame burning is kind of the canonical free energy chemical minimizing system it's dissipating literally the chemical potential energy and it's described perfectly by the dissipation of free energy into its environment and it's even that metaphor of like oh wait a minute if you put a candle under a glass jar it burns out the oxygen and then no more same thing happens with life forms that require oxygen to do respiration which basically is just taking that big drop in free energy that you can get from high energy molecules like fats and then just trickling that and yes making some heat but also using it to structure other macromolecular systems internal and external that allow other fats to get eaten yeah blue so i want to push back on dean for a minute um he's like got me stuck in this like temporal like i don't know zone so you were talking a little bit about um the relationship like a rock has now and iraq has net then but but i was not clear i want you to clarify i'm gonna really like push hard on you i want you to clarify what you mean uh like what is the difference there between a rock and something that's alive is it a relationship between now and then be it past or future or is it awareness of the past and the future and how do you know a rock is not aware or that a bacterium is aware so can you clarify like what actually is like this now then nest sure okay so so what i'm saying is i don't need i don't i don't believe if if you could i said i don't want anthro anthropomorphize it but if you could ask a rock does it need both the now and then i think the answer it would give back is i need one or the other i don't need both and to the point about the flame if i if i consider myself metaphorically to have it a fire burning inside of me if i'm an engine i have to know that in order to keep the fire going i have to bring in more energy i can't just like a candle burn myself out so i can't go unilaterally towards the conclusion i have to continue to bring up now and then oh i guess i'm getting closer to the end of my my energy cycle i have to now recapitulate and and bring myself back up to that place where i can continue to burn energy so that's what i'm why i'm saying i think if we want to look at this from the standpoint of of what what is actually using now and then versus what can continue to exist in it and i don't know if the word is entropic but a world of entropy that's one of those things that we could look at in terms of what free energy principle is asking us to incorporate in terms of a relationship not just not just the relationship between two objects in space but the amount of time that's incorporated in that relationship the now and the then i don't i'm not i'm not asking us to pin down what exactly that is i'm saying let's hold that up as as as either continuing or not like i think that um iraq must have a past in order to exist in the now to you to you and your relationship with that rock well even for the rock to be itself it didn't just appear out of nowhere or did it maybe right yeah that's what i'm saying i don't think it has a cell because it doesn't have it well so so perhaps what is necessary what is necessary for something to be is a delta between now and then like a rock was a could have been a rock then in the past and could be a rock then in the future but but doesn't require any kind of delta between now and then like any kind of change at all in its structure although it does change right i mean like pieces of it fall off and it crumbles away and turns to dust et cetera entropy right yeah right right so i mean it does change over time but but it doesn't require that that change and so so perhaps that delta is one of the things like i need to be different now than i was in the past and i need to be different now than i'm going to be in the future in order to be alive maybe here's a a meme it made me think of eternal sunshine of the spotless mind it's like a book or whatever spot it's kind of like a particle we talk a lot about particular states so eternal sunshine of the partic particle-less mind like if you don't have a particular state which is the internal and the blanket states those are the particular states if you don't have the existence of a it's kind of eternal it's just like an empty field now that doesn't have to be a realist claim about like what was there before the big bang and or genesis 1 it can just be like in the model before a partition and a time scale has been applied there was no distinctions but then once you engage a particular partitioning then if it is going to persist through model time and then maybe more speculatively through real time it does have to engage at least in looking like it's making adequate policy selection so one rock we're going to be scratching our heads over for a long time but then if we had a field of rocks some of them would be at and then there was like um one side of the rock field had some corrosive acid and one part didn't like the rocks that did take the policy to be on the other side those would be the ones that would look like they were persisting and so when we kind of zoom out to the population level which is also what happens at the end of this paper when we look at the population level of a similar type of entities then we start to see the strategies that lead to persistence versus the ones that don't like when people think about the rock i don't know they rarely talk about other rocks and about differentiating which ones are more lifelike or less lifelike but rather when you only have one reference point you can't really have a delta dean yeah and that that's this is where i think we can we all of us that are looking at this free energy and active inference thing can actually see a difference between three rocks on the ground and a domain then now and then can be looked at two ways they were just three rocks on the ground and now they're stacked there's a now and then but that's different than the partition between us as living things and those things that have now be been reordered and that's what this paper is asking us to look at do we want to look only at the way that the rocks have been reordered or do we want to look at the now and then of our relationship to the rock if we can include both we get a completely different picture and of course we even used the rock example back in the day back in the dot zero yeah so here was sort of the two um life lines or trajectories because both the person and the rock got there there was how they got there or what it takes to get there for both so here is a restatement of the point of the paper which is our strange attractor the simple point i make in this paper is that the free energy principle is not concerned with the sufficient conditions of existence but rather with what must have been the case necessity given that you exist it's not about figuring out what it takes to be alive it's about figuring out what took you so here's the lifeline of another system and um schrodinger's question at least as it's framed is kind of like asking you know give me the binary is it alive or not but that is predicated on us thinking we're alive i think that's fair to say it's not stated in there that humans are alive and that's my reference point or i'm alive or the writer of this article is alive but that is a very different question so figuring out what took you there is a different question than asking whether the past and the present the future of the rock is alive both entities to exist in a dissipative world will have done something like minimized free energy whether with respect to a complex boundary condition and a complex policy space like a human a simpler boundary condition and potentially a negligible afford in space like a rock like it's almost like people talk about where the rocks are alive but then are we alive and we should have a little bit of an answer there or what is alive what scale are we alive at so can i add to that daniel so if we look at some lots and lots of those those figures and we see t and then we see t plus one and then we see t plus two we can look at that two different ways now those are then and then and then in a linear way or we can look at it as then as an in relationship to now and again that's what i think axel is asking us to contemplate is both if we look at it simply as then and then i could be a rock but if i'm now relating it to some other time point that's when whatever whatever else free energy principle speaks to it says it isn't just another point on a continuum it's in a relationship with something else and i think that's what makes it really interesting in terms of the entanglement question what's necessary and what's because you can actually look at it as as another point on a continuum and it actually kicks out as sufficient but if you look at it as it a to for example now it becomes necessary and that's the i think going too far off on the tangent that's kind of where it gets really interesting it's how you are looking at it because there's a you asking how did i get here so i modified the slide a little bit now the living red line creature so we're just gonna say alive like me that's the kind of one that matters in the sense that it has a quasi-crystalline aperiodic information structure and it's negantropic those are the two answers that schrodinger provides in 1944 what is life we just can't ignore this is where we're looking through our lens and our perspective on the system so it's not like we get a sort of transparent look at or an objective or a monolithic look at the rock it's through our perception of the past present and that's where like the specific time scale that we choose to model it at because over if you speed up the rock it's just going to be burning like the candle now maybe that takes like millions of years just in as the surface reacts with oxygen and gets oxidized but isn't the candle oxidizing and then weren't we just talking about how we were oxidizing and then maybe the special part of our is that we're doing like this or something something like that like back casting with our metacognitive model how we got here well it's been this many years well since what since the hospital signed your birth certificate or since you were conceived as a zygote or conceived as an idea or what exactly that starting point and i think that's kind of why the starting point here is off the edge because again we may never know what is sufficient and sufficiency may include everything but when we talk about necessity we can actually do counter factuals we can do loss of function experiments necessity can be explored for complex systems in a way that sufficiency may never suffice like you were talking about since you were you know born conceived since you were an idea but what about like since you were recycled from like previously existing organic matter like like i mean literally we are all stardust right so like there's somehow some kind of partition gets created around us like at conception at birth at when right like so this whole temporal thing is is very interesting and it's like um you know would you exist could you exist without a mother or a father or someone to care for you as an infant like no um but i mean and that that distinction is um very different for different life forms like bacterium they don't need any care um they just but but is perhaps like the reproduction like the act of even binary vision is that sufficient care you know like what does that that relationship um look like in life and in aliveness and um you know we talk about like one of the necessary um qualities of of life or or you know i mean i'm not sure we can look at if have they changed but like the ability to to reproduce oneself is one of the the necessary um qualities for for being alive and um lots of things we produce that are not considered alive like i mean lots of things so um viruses nanobots uh you know i don't know it's interesting sorry i'm like off on a random tangent yeah thanks blue dean blue i don't really think that's that's tangential i think without without collapsing to like first year philosophy classes the fact that we can hold up two things as important one of the things that's important is which came first we spend a lot of time trying to figure out order through that lens the second lens is there's a there's a here and now and a it doesn't matter which came first it's just that we understand that the two things are different that's the second lens open they don't they don't play well together because the the one that wants to figure out the order is literally trying to find a source the other is just saying we're not going to collapse to one we're going to hold up two things as long as we possibly can they're bad playmates we shouldn't put them on a play date because they're inevitably going to fight with one another but if we have both of our eyes open what does that say then about us being able to get a better grip so that we can let go when free energy principle is included or incorporated in these these sloped moments because i think it's the two the figuring out which came first and here's the relationship between the two it's not one matters more than the other that creates the actual slope that's where as i said i don't want to turn it into a philosophy course but i don't think that's a tangent i think that's what we're trying to reconcile because we called it attention we can call it a you know they don't they don't play well together whatever but the bottom line is is that as long as we understand there are multiple arrows sometimes we can break them up as then and then and then and then and then and sometimes it's about the now and and i think that's the part that for for most of the stuff that we've seen in terms of how it's represented it's as daniel pointed out it's that spontaneity of partition part that we have to also include and if we do doesn't make it any easier they still don't get along but at least it tells us what's what what might be at play that's why i keep coming back to play it's uncertain who's going to win out in any particular moment well i think like while we're discussing when i think when was the partition created it is is important and like you know i mean in something like binary fission it's pretty easy to see you know like when the two you know cells separate and become you know by a certain distance by any distance that's that's when you know the partition is is made and perhaps that's true for all of the reproducing things or or perhaps not i mean i don't know like the chicken is separate from the egg right going back to the chicken and the egg but they separate very early so like the the you know maybe the separation physical separation isn't sufficient um like at that moment you don't have your own partition because the the baby chick or the egg could not fully develop without the mother chicken sitting on it even though they're physically separated so so it's this really sticky situation like where when is the partition made that makes someone an individual and that's why i always go back to that david crack power paper like when there's a bi-directional information flow upward and downward and that that's like what makes you um an individual and so like i don't know that um an egg a developing egg has that bi-directional information flow yet like maybe there's no downward causation in the egg there's no ability for the egg to act on its own other than do anything except develop like the capability for action or like the the downward causation like an egg is still in this emerging it's still going through emergence but you have to have both like you know emergence and downward causation for for individuality to exist i always go back to that so so i don't know in a rock is there downward causation or emergence like either one of those things do those exist and so that really that paper like just nailed it for me for for what um you know i don't know that it's necessarily the answer to schrodinger's but i think um it does a good job of getting us somewhere and i completely agree and and i guess what i'm suggesting is entanglement is a feature not a flaw and and we can enter and in that entanglement feature set we can look at order right and which came first and we can look at relation and so i know that if if there was an entanglement even though it's it's it's multiple light years away that entanglement tells me what the spin is and it's not the same direction there it's it's an opposite it's a polarness right so i agree with you and i think that's the feature here the feature here isn't we have to we can only do one or the other it the reason why we're even speaking to this is because it has to be both in order for that slope to eventually model out like i'm a big fan of model at the end i think that's way more sophisticated because i'm prepared to let the order and the relationship hang for a while breathe a bit do some back and forth scrap ufc punch one another and then i i get my model at the end now i understand why model at the beginning but i don't think that's the only way to go i think if we can do model at the end that's even more sophisticated because we have included more it's complexity in play let's return to now v3 of our figure these the colors are subject to change all right but red is realism blue is instrumentalism our favorite statistical instrument is the actin process theory which is a corollary of more axiomatic more general less applied free energy and then this is another system and the system of interest in this paper was a bacterium whether were structural realists or pretty much any stance other than the most hardcore denialism we exist like if we're not alive then what is the conversation to be had but when we ask how did i get here which is really happening or when we look at another system that's also really happening those are real events but both the reflexive question and the exteroceptive scientific modeling question are going to get filtered through instrumentalism i don't know if there's some way out of it's kind of like the difference between saying well you can escape plato's cave but then can you escape like using your eyes or using your generative model it's like a whole nother level of cave we have like active and the free energy principle this classic figure organism and the environment and then that is shown in the case of a brain as well as in a bacillus that particular partitioning about how we can think about the perception cognition in action and impact of autonomous systems once you're past the veil you're all within your instrumental model so that's where the question like well what were those priors in my bayesian bacterium and then the future is unknown because we can say well in my model the bacteria is minimizing free energy or i'm talking about this bacterium and i think physical systems minimize free energy it's a physical system so i think it's going to minimize free energy but as we established in this paper free energy minimization isn't sufficient to be alive it's not sufficient to persist because the candle is minimizing free energy too so we don't know just because it has existed and is minimizing free energy that it will exist in the future so that's kind of the unknown and then once we're post-partitioned we're all in the instrumentalism world even the framing of what is life it's an implicit realist playing field versus how do we model life the answers that schrodinger provides look for the information rich according to your information theory quasi crystal structure or look for using your thermometers and your heat sensors look for systems that appear to be locally creating order but then creating disorder surrounding them i don't know if anyone really denies that those are good ways to study life or that those are heuristics that can be applied it's just like saying well your instrumentalism doesn't align with my realism but what is your realism and who told you about it and how did you find out about it what are you gonna do about it what instrument are you going to use to investigate it it's interesting how the paper shines a light on how the entailment problem is really linked to some of the biggest themes that we talked about this year with realism and instrumentalism because again the entailment problem runs through and what do you think dean yeah i would say that nails it because if you want to look at it as which is more important realism or instrumentalism fine but if you're looking at the relationship realism and instrumentalism and how they relate to that's going to make a fuller picture it doesn't necessarily make it a cleaner simpler picture in fact it makes a messier more complex picture oh well sorry there's a relationship between those two things get over it grow up time to grow a nice long white beard and and smile because guess what it ain't going away just because it's more complicated so i have a problem with like i don't know realism overall because like what is even real like i view the brain as just an instrument through which we you know act and infer like this is this is it and every living system in my mind has like this um you know kind of computational input output that they use but but it's an instrument that they use and like the idea of realism is very um icky to me it's so subjective like what is even real i just don't i can't buy into the notion of some kind of shared reality for anyone we all have different affordances different preferences like i don't even know if i see the same color when i see yellow that daniel sees or dean sees like well you know how i perceive yellow can be totally different from how you perceive yellow and and that's why we all have different colors like favorite colors right with different colors that we like more and like less so you know i mean and people are color blind and so we know that it's just not that way so i don't know like it's all instrumentalism in my world like we're instruments i think it returns to the unifying approach versus unified theory we've been really mired in the mix-up of realism and instrumentalism and it sometimes is very slippery like how things are but then that's the twist with the theory of everything and all the free energy principle for a particular physics and all of that it uses sometimes even some of the same terms like thing but then by thing within the act infontology we might be referring to this particular thing so the blanket and the internal states and this particular thing is a cognitive model of and then could we have a model of those kinds of things maybe but could we have a model of real things or would this always be a sort of blurry discussion because everybody would be talking about their model anyway and it's also a lot like um the layers that existed in 32 which we'll return to in two days where there was the lorenz system that's like the realist system now it's one that we defined but it's really that way the time evolution of the lorenz system truly is defined by those equations that were granted to it so that's sort of the realism claim and then there was a first and a second order approximation to it which are also things but they got cleaner and cleaner and more tractable as we instrumentalized the thing we were looking at the question is always well how much of the attributes of the real system does the instrument retain do we retain false positives like kind of things about the territory that end up not helping our map do we have false negatives like we simplified the map a ton but we ended up missing really important features of the territory and if the territories are complex as we suspect they are to have persisted this way in a negentropic world well if they're complex how will we thread the needle with the appropriate coarse grainings as not to include false positives and false negatives in our map because just having the full description of the system is not going to be useful anything simpler we're going to be at super risk of dropping features that matter and so 32 showed that it was possible just at the mathematical level for example to make an approximatable laplacian approximation so tractable laplacian approximation but still retain the lyapunov exponent characteristic of a chaotic system so we we abstracted or we made approximations too but we still retained one of the essential features of it's going to be harder to do that to a bacterium than a lorenz system because the reality of the loren system is just numbers whereas the reality of bacteria is something probably different but it's a similar kind of question maybe we have some of the early versions of the tools that we would at least want to have in the future dean yeah and i'm not defending either realism or instrumentalism but i think in this paper and that paper which we're going to go back to if we if we use real or realism what they both point to is we can get it really right or we can get it really wrong or some degree of that i think that's where realism as a as a concept comes in it's not an absolute it's not a truth it's the variable as opposed to the instrument the instruments is trying to find out the precision and reality is about imprecision so again it's the relationship between those two things i'm not looking at realism in the classic definition of how other people see it i'm comparing it to the instrument which is trying to gain the precise and then so so what's the not precise okay i'll just i'll just throw that under realism for now that's how i square it so so dean we're on the same page then because like accuracy is relevant yeah exactly exactly so i i don't think you and i are apart or even remotely you know distant but it's how you see that relationship if you if you frame it one way something kicks out and if you frame it a different way something else kicks out and i'm not going to tell people who frame it the other way that they're wrong they're relatively safe in in their frame but it's got to be two frames at minimum otherwise we're we're lost we're a stone per se we're just going along for the ride and i don't know that that's i don't know that that's fair to stones well i don't know that we have any other choice but to go along for the ride honestly i mean and this goes back to you know the discussions that i've had about free will and um i discussed this with mike levin and i i mean really like you know if if we are just the sum of our generative model like if that if we that's just us we have no choice but to act accordingly unfortunately or or fortunately for us blue how's your i just the confusion's built in sorry i just wanted to say i think the confusion is built in i think the uncertainty part of this is built in we know that we're eventually our demise is near we just don't know when right and that's why i keep coming back to the where where did how did i get here is essentially asking and that goes back to way back when we were talking about people with um alzheimer's and in the hippocampus integrating where and when yeah and the the grid cells and the mirror neurons and all kinds of ideas that people have frames that are instrumental takes that blur the line because they're about real systems but then bring a little bit of an actin flavor to it i have one note then a question so here here's the red living system now it's shrunk and turned into an image and then here's here's the orange person studying bacteria so now it's like the blue and the orange person they can be having a conversation through the scientific literature or on a live stream or whatever like this is where we start to go from that rock is it alive or not to here's a population of rocks which ones are succeeding at being lifelike versus failing in terms of their dissipation so now here we have a minimum two people looking at the bacillus each with their own autoreflexive timelines and maybe the orange person doesn't like active that's fine they're using some other framework but they're not breaking out of the instrumentalism trap they're just doing some other instrument and then they're looking at the bacteria through their lens and then there's a conversation to be had but it's almost like the more layers the more super systems you add here it starts to be like i don't know less and less about the bacteria and oh well in your culture you were raised to think this way about this kind of scale or this kind of system and this is an interesting conversation but it's still nucleated around that bacteria and the model and then blue my question for you was over 21 how has your feelings on free will changed so like you know after i i'd say like the real the real um like mindset change shift happens more like at 25. um at least for me but um i don't know like you know i think that the whole experience of being a teenager is like identity seeking and putting yourself under the illusion that you have free will or you have like such a desire to have free will that that i i mean like that's what it is to be a teenager is like you go through this like identity crisis and then you liberate from the parents right and so i think that there's that like i definitely experienced that if that's what you're asking she already underwent the phase transition yeah i mean the phase like i already yeah i think um you know as teenagers we have like this illusion of and desire for freedom that is um what we're supposed to have and feel at that at that phase in life in order to break free and and maybe the age is irrelevant i mean maybe in some ancient prehistoric cultures they felt that at 10. but the the point is is that the desire to you is it physical separation or or to solidify your own partition right like um the desire to be your own independent person so i think like i i thought that you know freedom and free will existed at that point and now like um i mean i feel like we're essentially enslaved by our generative model we're enslaved by the past and the future in the now i think that takes us to one of the last prepared questions we had one of the key words of the paper was historical science we talked in the dot zero about how there's all these historical sciences you won't see chemistry biology physics not those stem sciences but history and then all these other that draw data from records of past events as opposed to experimental or operational science but asked it then and asked it now aren't experiments in the past i don't see how merely having hypothesized and planned acquired data when you're analyzing it it becomes in so somebody who's studying heraldry could hypothesize oh in this library on this shelf in this book i might find something important and i'll analyze it and then that data is in their past the historical sciences are contrasted with maybe what we could call the future sciences so not like the future of science which we hope is awesome and participatory and accessible and whatnot but science is about the future which range from the sort of speculative and almost metaphysical to the extremely applied where do we sit now with historical and anticipatory or future oriented sciences in the now dean so i'd like to try and take a a first stab at that let's and let's use it let's use the free will question if we looked at it historically as bring out the scrolls this is what happened or we look into the future and oh this this person's thought a lot about what's going to happen tomorrow let's build the scroll for tomorrow that's one way of looking at the question of what is free and will if we look at as a relationship we could free will or follow will meaning i can't break free i'm i'm entrapped or whatever and then what we could do is we look at the relationship as an ongoing an ongoing exercise between free and follow now neither neither the free or follow are going to fall neatly and tidally into a scroll but they have a relationship whether we scroll it out as a historical thing or we scroll it into as a projecting thing but the ability to do both not just examine free will as a linear but also be able to relate it to so what's what is non-free will it's like will follow and then now compare the follow and the free give it kicks out a different model the one that's one that's messier one that's up for debate and i think putting putting both of them up on the shelf and examining both is to look at essentially i think that i i recently just learned that futurology is a science um which which was super fascinating to me um like the study of of predictive methods right and there's so many ways that that we you know have used to predict but um something that i thought was was really really interesting was um they used to before the internet was like trolled by bots you used to be able to take like message boards and chat boards and like you know essentially twitter streams um and i'm not sure how accurate this is anymore because there's so much like just artificially generated text but when it was humans creating the text they used to actually text mining as as like a study of the future like so like a prediction analysis based on text analysis so it's um and was very effective so i hear but i think that this like study of future predicting be it oracles or astrology or what have you is really kind of fascinating because you can use all kinds of things active inference for prediction or anything right so um i don't know i think it's interesting to compare and contrast the different ways and um the ways that if you're open to accepting insight from tarot cards or the iching or the oracle of delphi active inference or you know text analysis if you're open i think that all of the future or potential potential future avenues can can be very insightful like that's a possibility if only opening you up to the fact that that's a possibility i think that that that's the the real value in in futurology thanks blue dean yeah i think you're i think you're right i think the law of large numbers and the fact that we now have the ability to look at much larger data sets and and apply probability and statistics to those things should give us some sense of direction that we probably didn't have affordances to before and the messiness and this is why a lot of times philosophers and scientists are in that same tension of relating to is that still doesn't collapse down to i go and see my my tarot card reader and he gives me an answer and i believe because that's not a law of large numbers so now do i follow or am i free it's messy right so yeah anyway well and does history repeat itself right like there's the old adage history repeats itself can we use history oh can we use history to um predict the future that's where it's sort of mobius strips strange loops around when you make the model based upon the past that's anticipating the future and then you start acting that way going forward in the future it's almost like the timeline has folded because now the past is influencing the future in a in a way that might start to diverge from not like what it would have done obviously any policy is going to diverge from some counterfactual but it starts to um potentially lose its model adequacy events from the past whose context has changed like let's just say the correlation two cryptos we look at the past correlation but the context is changing so we're gonna have a model that's fit on their correlation in the past and then especially action on that but even without our action on that that relationship may change and i think that's something interesting about actin that it it puts perception cognition and action in the same loop always unfolding kind of like uda so that there always is the opportunity for flexibility as surprises come in and then also what you were saying about free will following well freeing and then to bring it back to realism and instrumentalism like here's the free will module in my model it's like here's how it works here's what it does here's the inputs here's the outputs so does the thing have free will yep in my model it does what about the actual thing i thought i told you we stopped talking about that a few years ago or something like that like we're not talking about that or that's an interesting question but now we're at square zero with what is real instead of i made a hypothesis about free will module that's just what i'm calling it that's not a free will module that's a random number generator or that's something that is doing its own prediction and then makes a decision based upon something else or that's like a preference vector yeah and again today i don't i don't think it matters i know most people would say it does but i don't think it matters whether you start with a slide that has a blue silhouette looking at a gray rock or you start with a slide that seems appears to be much more more um that's what's the word for it much more further along in terms of orange lines you can go both directions yay that's the thing that we should be celebrating like it may appear from a from a constructivist point of view which came first that there's no way that you could have started with the orange and ended up with the much much less informational 10 but yes you could and we oftentimes do that's that's actually if you go from 13 to 10 in the slide deck that is free playing out yeah we could look at this one and we could go oh that's minimizing we're talking about instrumentalism so we just know that when we draw an arrow of some agentic system we know that we're talking about instrumentalism and we know that the instrument that we're talking about is free energy principle active inference so that's how you step back from here to here and then just go okay so and then we know that models have their own internal time and that that's not the same as like real time so let's just simplify and then let's just a little bit less academic not worried about which paper i should write let's just focus on the motivation of that paper is it alive it's like we step back i don't know is that it's like backwards and forwards in kairos as we move forward in kronos dean and that's where instructionalism and i forget what that it was a hippolyta paper i think yeah interactive construction performance where the action is that's where this that's where those four slides prove out her paper or at all paper because the reality is it's not simply constructivism it's disentanglement there's there's a product at the end of it that appears to be a construction don't get me wrong however what actually generates four versions four iterations is a dis entanglement not an integration and we have to be able to hold up both if we're actually going to get to a place where we understand house understanding emerges and that's where i that's where i find this stuff really exciting because if it's about interaction here's the proof let's in the last few minutes where do we go so not where we came from [Laughter] or when do we go um mom are we there then blue will be working on some fun literature analysis the history of active inference literature and free energy principle half 22. we hope to see many new papers coming in from authors that we know as well as from new authors and new perspectives and selecting cutting edge papers and recent papers that are coming out and preprints ones that might not be done see where the live stream can insert itself into that literature stigma g maybe go back to some earlier classics for contrast and understanding how ideas change through time and where we are now who we are now all these other variants of the question that we talked about i mean we're gonna be doing it on wednesdays uh next year starting one hour later at uh i think 16 utc at 8 am pacific the weekly paper discussions or just any other system of interest where are we going or what will we do differently in 22 i'm curious to see we're going to have our 40 second or 40 second live livestream that's gonna be epic we have to like choose wisely choose choose a paper wisely and also like our first paper of the year is gonna maybe set the tone for what we're gonna you know learn and do and be about this year what was our first paper last year in 21 daniel it was um adam saffron and colleague 13.1 let's look on the coda site um oh we haven't updated it yet but it was 13. on the cybernetic big five oh that was epic too that was really good and actually like you know the big five was that was my first time learning about it and it was kind of we covered all of those this year uh several times so we'll have to have to similarly choose like an all-encompassing first paper hopefully we go into more participation more expansion and more broad acceptance or recognition of the fep i hope that's where we go that would be cool do we have any other comments or we can give sort of a yes dean please one last thing i think i didn't come in i wasn't there a year ago but one of the things that has recently happened is that we've introduced another walnut shell the potential for a 0.3 which i actually think for most people would be like yeah okay so what you're gonna you're gonna stare at your navel longer but i actually think it's a really exciting all right pete dean yeah please from the from the oh wow all right dean yeah but the doctor affordance is cool so that's a really good point dean um the walnut shell the navel gazing the dot three continue yeah i just was gonna say i think it's it's actually a really exciting time because when we add another shell we're actually practicing the active inference instead of just talking to it in different versions and that that to me is if next year means that there's reveal more q unknown and we actually have to play with that uncertainty space as opposed to walking into it and saying well we've now had this many live streams so now we could no we should actually be doing more shells 1p and more options hidden that's when it gets really exciting so that's what i'm looking forward to so that's a super good point dean and so if people have other live streams that they wish that they were a part of or you know other authors want to come back for a revisit to do a dot 3 of any live stream that they've been on before like reach out we always have the affordance to make an anything.3 so yeah like we can revisit old things and move on to new things and um do both of them together history the weird folding right yeah we can do guest streams all kinds of fun stuff and then also just so people know um if you're ever part of the live stream you'll see this slide asking you to go through a checklist this is our 21 version and we hope people go through it i think sometimes they do lots of information on participation on things to check before you start if you're a participant so not like the author of the paper um but just somebody who's gonna be hanging out and giving any thoughts on a paper there's some checklist for you for a presenter like an author who's gonna be presenting a talk there's information for them there's also the broadcast role so that's the person with a stable internet and a computer that can run a streaming software there's a conversational facilitator yeah exactly dean's cabin dot stream um a little stream through a tin can then there's the organizers checklist and then we're always growing and changing as active inference lab because we're a participatory lab and that includes every single possible role including many that are not listed here if it's something that you're interested in and if you're listening and your regime of attention has extended this far maybe you'd like to join a live stream or make a suggestion or play one of these other roles because it is a multiplayer game and i hope we're having fun when we get to do it so dean and blue and steven from earlier and dave from earlier as well today always appreciated it was an awesome year i think we all learned a lot by doing and updated our generative models thanks daniel really appreciate it thanks blue fantasy